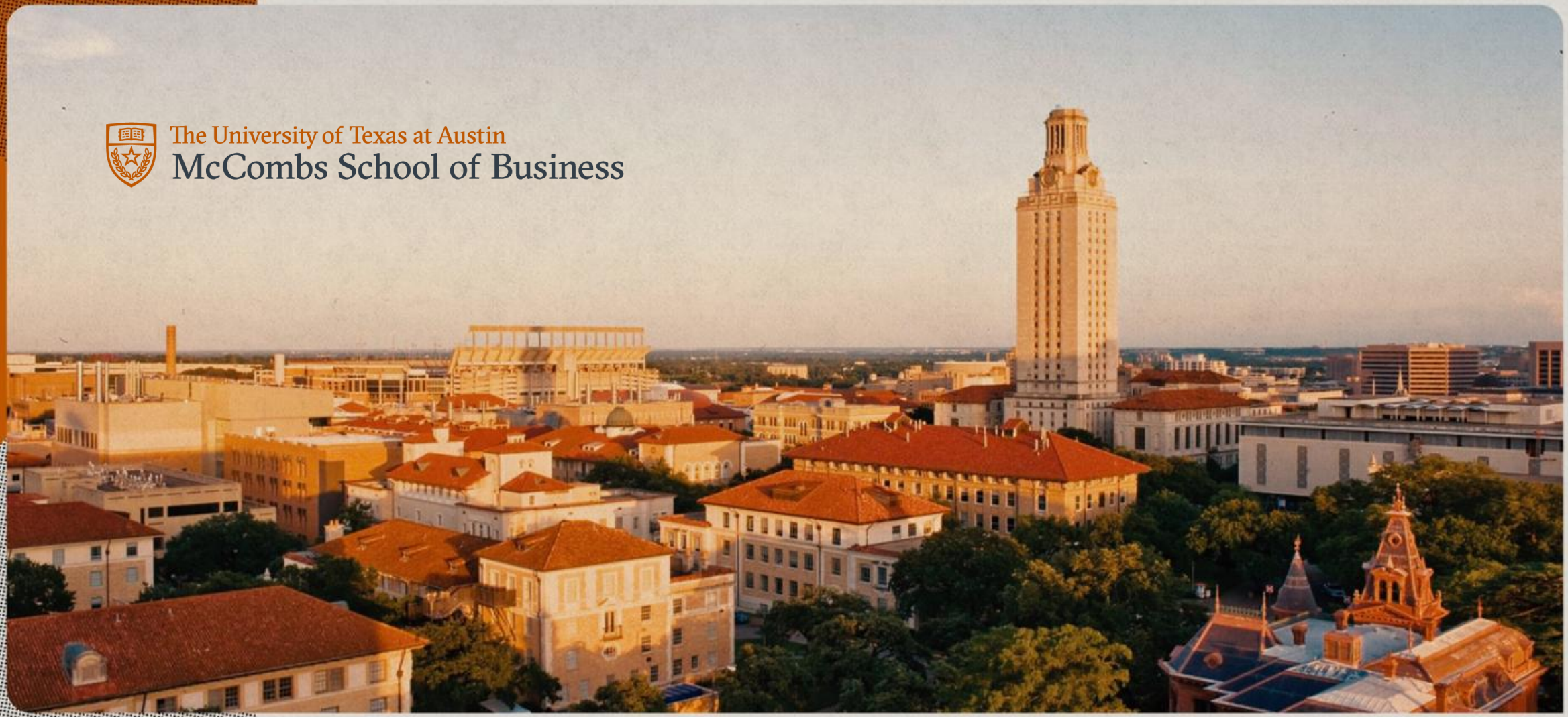



# D9: Gaussian Mixture Models

# GMM from 40,000 feet

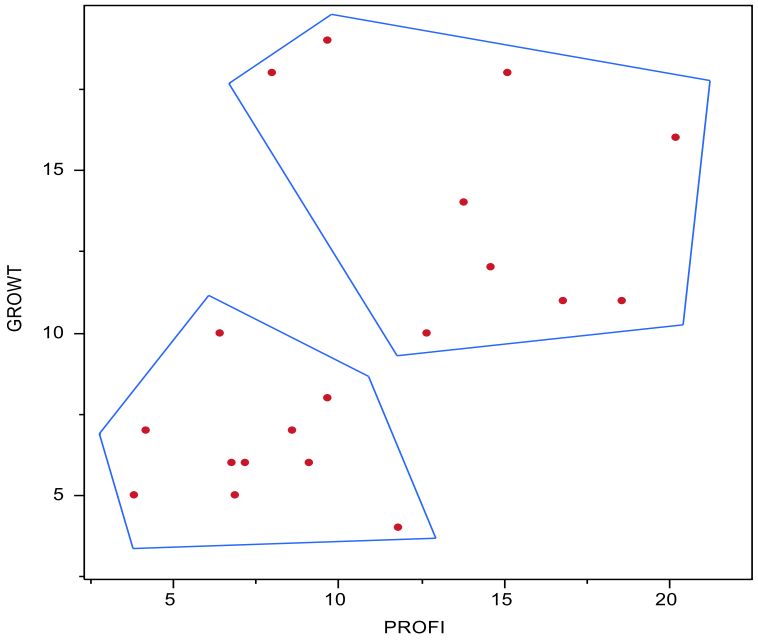
**GMM** = **G**aussian **M**ixture **M**odels

The GMM approach to cluster analysis views the data as having been generated from several multivariate normal distributions, mixed together in various proportions. Each distribution  generates a different cluster. Unlike K-means, which assigns each point to a single cluster, GMMs assign probabilities to each point for belonging to each cluster. This allows GMMs to model more complex data distributions and handle overlapping clusters more effectively.

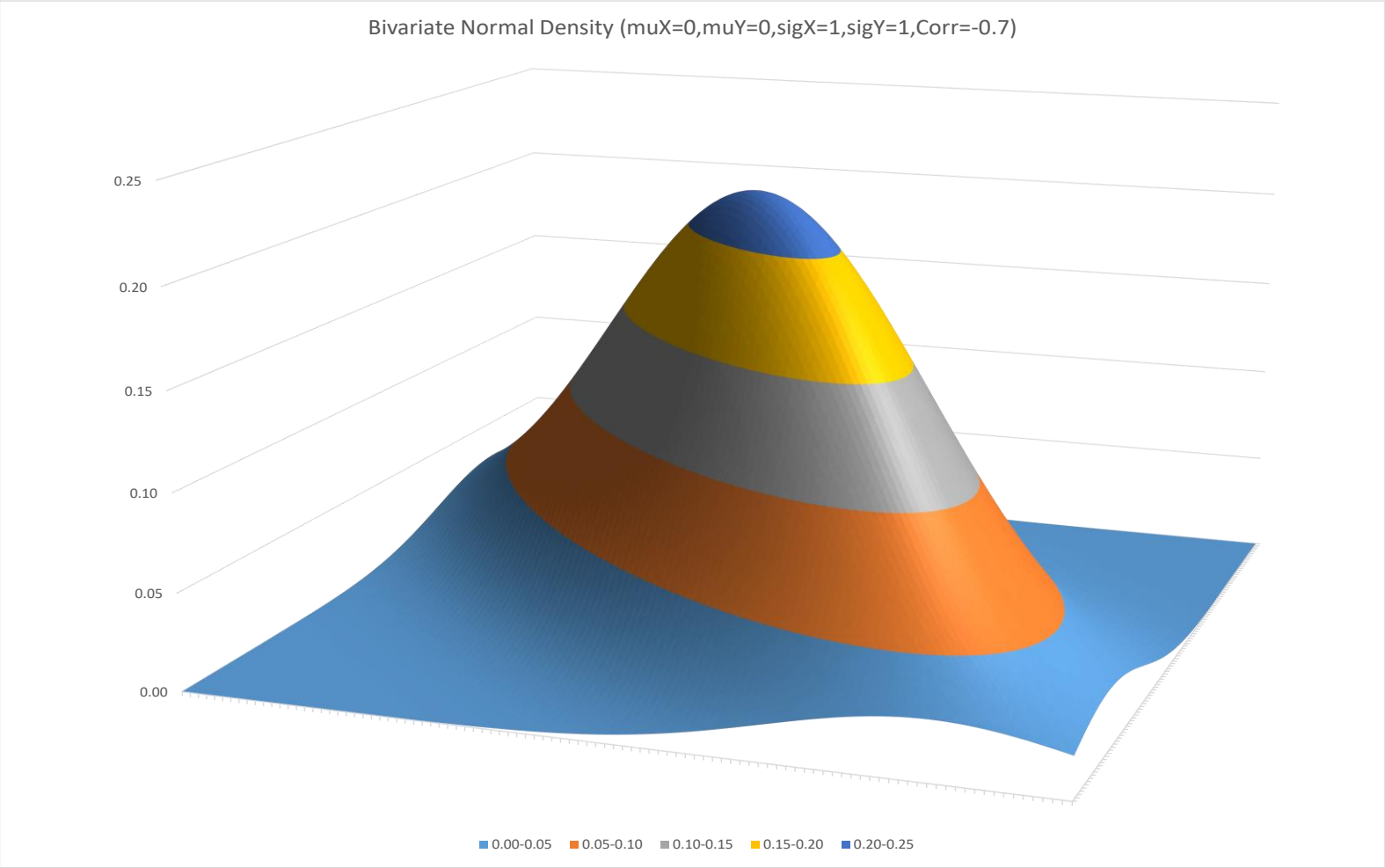
# GMM from 40,000 feet

**GMM** = **G**aussian **M**ixture **M**odels

Ex: From CA notes:  Assumed two bivariate normal distributions mixed in approximately equal proportions.



# Example of Bivariate Normal Distribution



# GMM Cluster Analysis

Theoretical foundations:

- **Means** ($\mu_k$): The center of each Gaussian component.
- **Covariances** ($\Sigma_k$): The shape and orientation of each Gaussian component.
- **Mixing coefficients** ($\pi_k$): The weight of each Gaussian component in the mixture, with the constraint that the sum of all mixing coefficients equals 1.

The probability density function of a GMM with $K$ components is given by:

$$P(x|\theta) = \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(x|\mu_k, \Sigma_k)$$

where $\mathcal{N}(x|\mu_k, \Sigma_k)$ represents the Gaussian distribution with mean $\mu_k$ and covariance $\Sigma_k$.

# GMM Cluster Analysis

Theoretical foundations:

**How GMM Works**

1. **Initialization**: Initialize the parameters ($\mu_k$, $\Sigma_k$, $\pi_k$) randomly or using a method like K-means.
2. **Expectation Step (E-step)**: Calculate the probability that each point belongs to each cluster using the current parameters.
3. **Maximization Step (M-step)**: Update the parameters to maximize the likelihood of the data given these probabilities.
4. **Convergence**: Repeat the E-step and M-step until the parameters converge.

The EM (Expectation Maximization) algorithm is applied by iteratively performing the E-step and the M-step.

# R implementation of GMM clustering

Install the package mclust:

```
install.packages("mclust")
library(mclust)
gmm <- Mclust(data)
```

In default mode there are no parameters to specify.

All variables in *data* will be used to cluster.

# Applications of GMM Clustering

Widely used in various fields b/c of flexibility and ability to model complex data. Some examples:

**Speech Recognition**: Modeling the distribution of speech features for different phonemes.

**Finance**: Identifying different market regimes or clustering financial time series data

**Image processing**: Segmenting images into regions based on pixel intensity distributions

# Limitations of GMM Clustering

**Assumption of Gaussian Distribution**: GMMs assume that the data is generated from a mixture of Gaussian distributions, which may not always be the case.

**Computational Complexity**: The EM algorithm can be computationally intensive, especially for large datasets or high-dimensional data.

**Sensitivity to Initialization**: The performance of GMMs can be sensitive to the initial parameter values. Poor initialization can lead to suboptimal clustering results